MH1100 Calculus I

Handout 1 - Functions

Synopsis

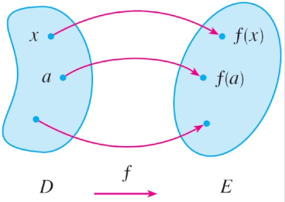
Overview of functions, and examples.

- Function, domain, range, domain convention
- The graph of a function, the vertical line test
- Piecewise defined functions, even and odd functions, increasing and decreasing functions
- Polynomials, power functions, rational functions, algebraic functions, trigonometric functions, exponential functions, logarithmic functions
- Transformations of functions, combinations of functions

What is a function?

Definition

A function f is a rule that assigns to each element x in a set D exactly one element, called $f(x)$, in a set E.



Range (f) $= \{ f(x) \in E \mid x \in D \}$

Definition

The set D is called the domain of the function. The range of f is the set of all possible values of $f(x)$ as x varies throughout the domain.

not necessarily every element in E

Ways to represent a function

In Calculus I we will mainly study functions for which the sets D and E are sets of real numbers.

Ways of representing a function

There are many different ways of describing functions. For example,

- verbally (by a description in words)
- numerically (by a table of values)
- visually (by a graph) → sometimes easier to use graph to understand.
- algebraically (by an explicit formula) → most of the time we use this

Any description which gives you enough information to unambiguously calculate the value at every point of the domain is OK (logically correct).

Visualizing a function

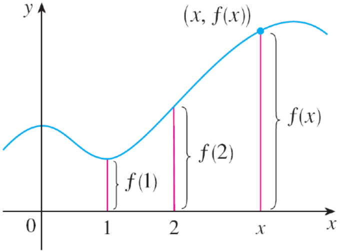
The graph of a function

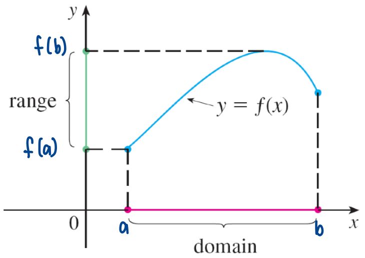
The most common method for visualizing a function is its graph.

If f is a function with domain D, then its graph is the set of ordered pairs

$$\{(x, f(x)) | x \in D\}.$$

In other words, the graph of f consists of all points (x, y) in the coordinate plane such that $y = f(x)$ and x is in the domain of f.



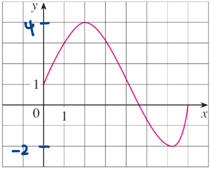


Function examples

Example 1

The graph of a function f is shown below.

- Find the value of $f(1)$.
- What are the domain and range of f?



Solution. We see from the above figure that the point $(1, 3)$ lies on the graph of f, so the value of f at 1 is $f(1) = 3$.

We see that $f(x)$ is defined when $0 \leq x \leq 7$, so the domain of f is the closed interval $[0, 7]$. Notice that f takes on all values from -2 to 4, so the range of f is $\{y| - 2 \leq y \leq 4\} = [-2, 4]$.

Function examples

Exercise

If $f(x) = 2x^2 - 5x + 1$ and $h \neq 0$, evaluate the **difference quotient**

$$\frac{f(a+h) - f(a)}{h}$$

$$\frac{f(a+h)-f(a)}{h} = \frac{2(a+h)^2 - 5(a+h) + 1 - 2a^2 + 5a - 1}{h}$$

$$= \frac{2a^2 + 4ah + 2h^2 - 5a - 5h - 2a^2 + 5a}{h}$$

$$= \frac{2h^2 + 4ah - 5h}{h}$$

$$= 4a + 2h - 5$$

When $h \to 0$

$= 4a - 5$

$f'(x) = 4x - 5$

Function examples

Example 6

Find the domain of each function.

(a) $f(x) = \sqrt{x+2}$ (b) $g(x) = \dfrac{1}{x^2 - x}$

Solution

- Because the square root of a negative number is not defined (as a real number), the domain of f consists of all values of x such that $x + 2 \geq 0$. This is equivalent to $x \geq -2$, so the domain is the interval $[-2, \infty)$.
- Since division by 0 is not allowed, we see that $g(x)$ is not defined when $x = 0$ or $x = 1$. Thus the domain of g is

$$\{x | x \neq 0, x \neq 1\}$$

$x^2 - x \neq 0$
$x(x-1) \neq 0$
$\therefore x \neq 0, x \neq 1$

which could also be written in interval notation as

$$(-\infty, 0) \cup (0, 1) \cup (1, \infty).$$

Function examples

Domain convention

If a function is given by a formula and the domain is not stated explicitly, the convention is that the domain is the set of all numbers for which the format makes sense and defines a real number.

Exercises

What are the domains of the following functions?

- $\{x \in \mathbb{R} \mid x \neq 2, x \neq 3\}$
$$f(x) = \frac{1}{x^2 - 5x + 6}$$
- $\{x \in \mathbb{R} \mid x \neq k\pi, k \in \mathbb{Z}\}$
$$g(x) = \frac{1}{1 - \cos^2 x}$$
- Domain of (h) = ϕ → emptyset, not defined.
$$h(x) = \frac{1}{\sqrt{x-1}} - \sqrt{1-x}$$

1. When $x^2 - 5x + 6 = 0$
$(x-3)(x-2) = 0$
$x = 2$ or $x = 3$
$\therefore$ $x \neq 2$ and $x \neq 3$
$\because$ $x^2 - 5x + 6 \neq 0$

2. $1 - \cos^2 x \neq 0$
$\cos^2 x \neq 1$
$\cos x \neq \pm 1$
$x \neq k\pi$ (k is an integer)

3. $1 - x \geq 0 \Rightarrow x \leq 1$
$x - 1 \geq 0 \Rightarrow x \geq 1$
$\sqrt{x-1} \neq 0 \Rightarrow x \neq 1$

More about the graph of a function

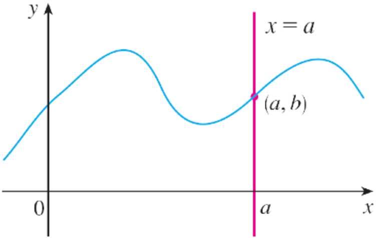
The graph of a function is a curve in the xy-plane. But which curves in the xy-plane are graphs of functions?

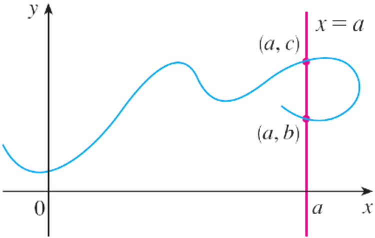
Must strictly only intersect ONCE!!!

The vertical line test

A curve in the xy-plane is the graph of a function of x if and only if no vertical line intersects the curve more than once.

This is not a graph of a function.



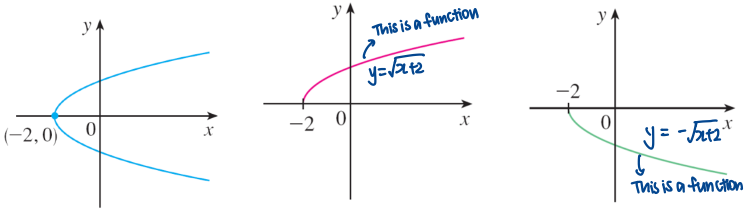


More about the graph of a function

The parabola example

The parabola $x = y^2 - 2$ shown left below is not the graph of a function of x because there are vertical lines that intersect the parabola twice.

$x = y^2 - 2$ is not a function of x, but is a function of y.

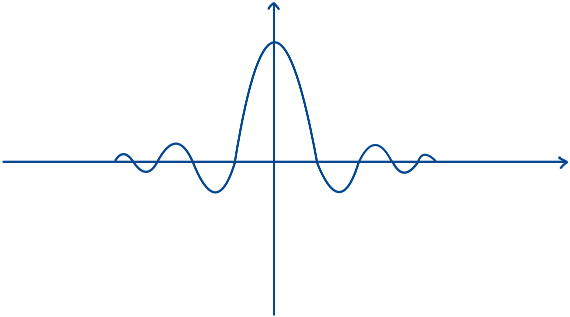


The equation $x = y^2 - 2$ implies $y^2 = x + 2$, so $y = \pm\sqrt{x+2}$. Thus the upper and lower halves of the parabola are the graphs of the functions $f(x) = \sqrt{x+2}$ and $g(x) = -\sqrt{x+2}$.

Exercise

Draw the graph of $f(x) = \frac{\sin x}{x}$.

$\mathrm{Dom}(f) = \{x \in \mathbb{R} \mid x \neq 0\}$



Piecewise defined functions

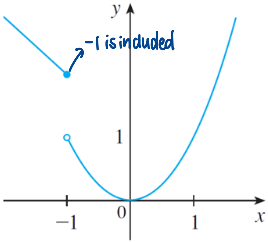
The functions are defined by different formulas in different parts of their domains. Such functions are called piecewise defined functions.

Example 7

Sketch the graph of the function defined by

Dom(f) = $\mathbb{R}$

$$f(x) = \begin{cases} 1 - x, & x \leq -1 \\ x^2, & x > -1 \end{cases}$$



Piecewise defined functions

Absolute value function

The absolute value function $|\cdot|\colon \mathbb{R} \to \mathbb{R}$ is the function defined by the rule

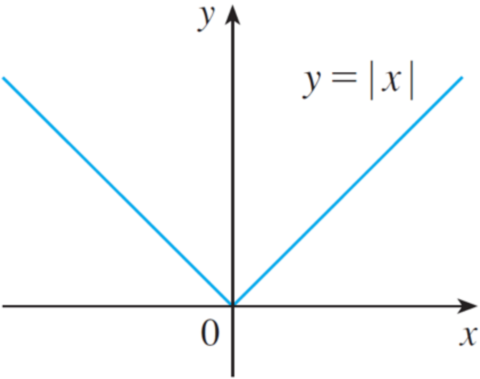
$$|a| = \begin{cases} a, & \text{if } a \geq 0 \\ -a, & \text{if } a < 0 \end{cases}$$

Note that the absolute value of a number a, denoted by $|a|$, is the distance from a to 0 on the real number line.

Piecewise defined functions

Example 8

Sketch the graph of the absolute value function $f(x) = |x|$



$f'(0)$ does not exist, it is not defined.

NEED HELP!

Exercise

Draw the graph of the function $f\colon \mathbb{R} \to \mathbb{R}$, which is defined by the rule

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is rational} \\ 0, & \text{otherwise} \end{cases}$$

Dirichlet Function

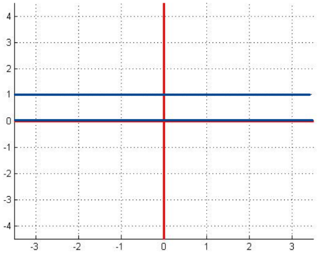
$$D(x) = \begin{cases} x, & x \in \mathbb{R} \\ 0, & \text{otherwise} \end{cases}$$

why wts

$x \in [-1, 1]$ huh???

Note: The rational numbers are the numbers of the form $\frac{m}{n}$, where m and n are integers and $n \neq 0$.

$f(\frac{m}{n}) = 1$



$\sqrt{2} \to$ irrational number

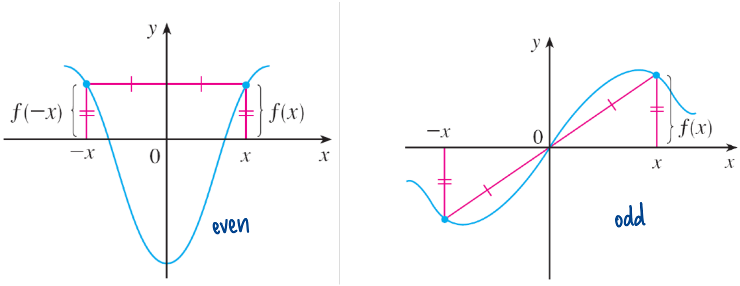
$x = \sqrt{2} + \frac{m}{n} \longrightarrow$ still irrational

$f(x) = 0$

Even and odd functions

Even function

If a function f satisfies $f(-x) = f(x)$ for every number x in its domain, then f is called an even function.



Odd function

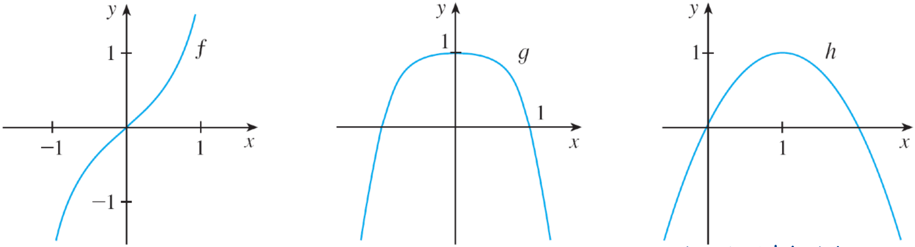
If a function f satisfies $f(-x) = -f(x)$ for every number x in its domain, then f is called an odd function.

Even and odd functions

Example 11

Determine whether each of the following functions is even, odd, or neither even nor odd.

(a) $f(x) = x^5 + x$ polynomial of order 5 (b) $g(x) = 1 - x^4$ polynomial of order 4 (c) $h(x) = 2x - x^2$



not symmetrical about the y-axis

Even and odd functions

Example 11 (Continued)

Determine whether each of the following functions is even, odd, or neither even nor odd.

(a) $f(x) = x^5 + x$ (b) $g(x) = 1 - x^4$ (c) $h(x) = 2x - x^2$

Solution

- $f(-x) = (-x)^5 + (-x) = -x^5 - x = -(x^5 + x) = -f(x)$.

Therefore f is an odd function.

- $g(-x) = 1 - (-x)^4 = 1 - x^4 = g(x)$.

Therefore g is an even function.

- $h(-x) = 2(-x) - (-x)^2 = -2x - x^2$.

Since $h(-x) \neq h(x)$ and $h(-x) \neq -h(x)$, we conclude that h is neither even nor odd.

Even and odd functions

Exercise

Let $f(x)$ be defined on the closed interval $[-a, a]$, prove that f can be expressed as the sum of an odd function and an even function.

$$f(x) = \overset{\text{odd}}{O(x)} + \overset{\text{even}}{e(x)} \quad \text{—} \ (1)$$

$$f(-x) = O(-x) + e(-x)$$

$$= -O(x) + e(x)$$

$$\therefore f(-x) = -O(x) + e(x) \quad \text{—} \ (2)$$

(1) + (2)

$$f(x) + f(-x) = 2e(x)$$

$$\boxed{e(x) = \tfrac{1}{2}[f(x) + f(-x)]}$$

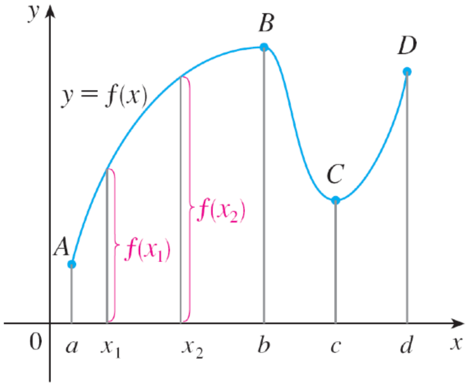
(1) − (2)

$$f(x) - f(-x) = 2O(x)$$

$$\boxed{O(x) = \tfrac{1}{2}[f(x) - f(-x)]}$$

Increasing and decreasing functions

The graph shown below rises from A to B, falls from B to C, and rises again from C to D. The function f is said to be increasing on the interval $[a, b]$, decreasing on $[b, c]$, and increasing again on $[c, d]$.



Increasing and decreasing functions

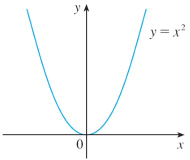
Definitions

A function f is called increasing on an interval I if

$$f(x_1) < f(x_2) \qquad \text{whenever } x_1 < x_2 \text{ in } I$$

It is called decreasing on I if

$$f(x_1) > f(x_2) \qquad \text{whenever } x_1 < x_2 \text{ in } I$$



$[0, \infty]$ increasing function
$[-\infty, 0]$ decreasing function
$[a, b]$ $a<0<b$, neither $\uparrow$ or $\downarrow$

In the definition of an increasing function, the inequality $f(x_1) < f(x_2)$ must be satisfied for every pair of numbers x_1 and x_2 in I with $x_1 < x_2$.

Increasing and decreasing functions

Example B

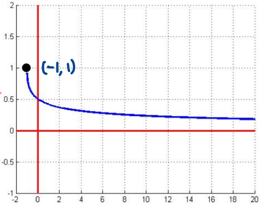
Find the domain and range of the function $f(x) = \frac{1}{1+\sqrt{1+x}}$

For $x_1 < x_2$ in $[-1, \infty)$

$$
\begin{aligned}
&f(x_1) - f(x_2) \\
&= \frac{1}{1+\sqrt{1+x_1}} - \frac{1}{1+\sqrt{1+x_2}} \\
&= \frac{\sqrt{1+x_2} - \sqrt{1+x_1}}{(1+\sqrt{1+x_1})(1+\sqrt{1+x_2})} \\
&= \frac{\sqrt{1+x_2} - \sqrt{1+x_1}}{(1+\sqrt{1+x_1})(1+\sqrt{1+x_2})} \times \frac{\sqrt{1+x_2} + \sqrt{1+x_1}}{\sqrt{1+x_2} + \sqrt{1+x_1}} \\
&= \frac{x_2 - x_1}{(1+\sqrt{1+x_1})(1+\sqrt{1+x_2})(\sqrt{1+x_2} + \sqrt{1+x_1})}
\end{aligned}
$$

$> 0 \therefore f(x_2) > f(x_1)$

rationalise the numerator



$f(x)$ always > 0

$f(-1) = 1$

Solution To begin, note that the expression $f(x) = \frac{1}{1+\sqrt{1+x}}$ makes sense precisely when $x \geq -1$. Thus, the domain of f is $[-1, \infty)$.

$1 + x \geq 0$

$x \geq -1$

Now we turn to the range. We can show that f is a decreasing function. Because f is a decreasing function, the graph starts at $f(-1) = 1$ and decreases as x increases. Thus, the range of f is $(0, 1]$.

A catalog of essential functions

Polynomials

A function P is called a polynomial if

constant

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$$

$n = 0, 1, 2, 3 \ldots$ $\because$ constants are polynomials → polynomial of order zero

where n is a nonnegative integer and the numbers $a_0, a_1, a_2, \cdots, a_n$ are constants called the coefficients of the polynomial.

The domain of any polynomial is $\mathbb{R} = (-\infty, \infty)$. If the leading coefficient $a_n \neq 0$, then the degree of the polynomial is n.

A polynomial of degree 1 is of the form $P(x) = mx + b$ and so it is a linear function. A polynomial of degree 2 is of the form $P(x) = ax^2 + bx + c$ and is called a quadratic function. A polynomial of degree 3 is of the form $P(x) = ax^3 + bx^2 + cx + d$ $(a \neq 0)$ and is called a cubic function.

A catalog of essential functions $2^{\frac{m}{n}} = \sqrt[n]{2^m}$

Power functions

A function of the form $f(x) = x^a$, where a is a constant, is called a power function. $x^2, x^0, x^{\sqrt{2}}$

still a constant

Rational functions

A rational function f is a ratio of two polynomials:

rational number

$\frac{p}{q}$ where $p, q \in \mathbb{Z}$ and $q \neq 0$

$$f(x) = \frac{P(x)}{Q(x)}$$

where P and Q are polynomials. The domain consists of all values of x such that $Q(x) \neq 0$.

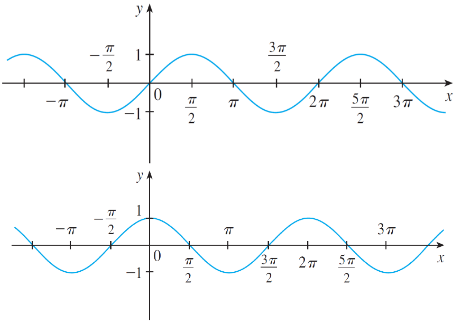
Algebraic functions

A function is called an algebraic function if it can be constructed using algebraic operations (such as addition, subtraction, multiplication, division, and taking roots) starting with polynomials.

A catalog of essential functions

Trigonometric functions

There are 6 trigonometric functions: sine, cosine, tangent, cosecant, secant, and cotangent functions.



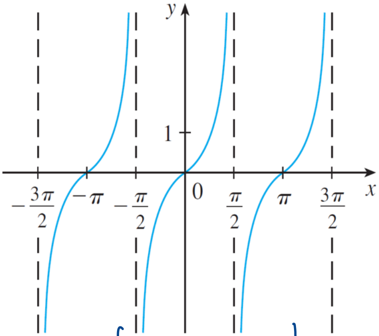
$f(x) = \sin x$
$f(x+2\pi) = f(x)$

$g(x) = \cos x$
$g(x+2\pi) = g(x)$

For both the sine and cosine functions the domain is $(-\infty, \infty)$ and the range is the closed interval $[-1, 1]$. An important property of the sine and cosine functions is that they are periodic functions and have period 2π.

Trigonometric functions (Continued)

There are 6 trigonometric functions: sine, cosine, tangent, cosecant, secant, and cotangent functions.



$h(x) = \tan x = \frac{\sin x}{\cos x}$

Domain of $h = \{x \in \mathbb{R} \mid x \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z}\}$

The cosecant, secant, and cotangent functions are the reciprocals of the sine, cosine, and tangent functions, respectively.

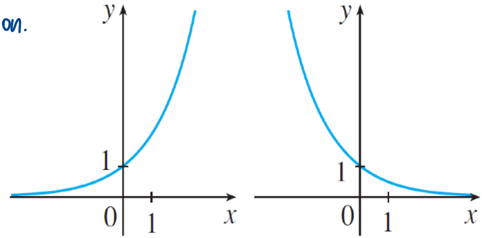
A catalog of essential functions

Exponential functions

The exponential functions are the functions of the form $f(x) = b^x$, where the base b is a positive constant. $0<b<1$ or $b>1$

The graphs of $y = 2^x$ and $y = (0.5)^x$ are shown below. In both cases the domain is $(-\infty, \infty)$ and the range is $(0, \infty)$.

When $b>1$,
b^x is an increasing function.
When $0<b<1$,
b^x is a decreasing function

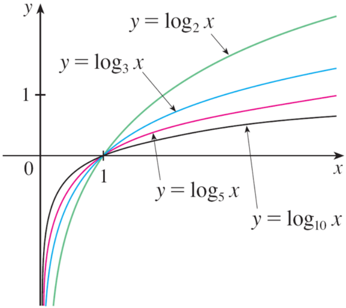


A catalog of essential functions

Logarithmic functions

The logarithmic functions are the functions of the form $f(x) = log_b x$, where the base b is a positive constant. They are the inverse functions of the exponential functions. $b>1$, increasing function $0<b<1$, decreasing function

The graphs of $y = log_b x$ $(b = 2, 3, 5, 10)$ are shown below. In each case the domain is $(0, \infty)$ and the range is $(-\infty, \infty)$.



Exercise

Classify the following functions as one of the types of functions that we have discussed.

(a) $f(x) = 5^x$ (b) $g(x) = x^5$ (c) $h(x) = \frac{1+x}{1-\sqrt{x}}$

(d) $u(t) = 1 - t + 5t^4$

a) $f(x) = 5^x$ is an exponential function. The x is the exponent.

b) $g(x) = x^5$ is a power function

c) $h(x)$ is an algebraic function

d) $u(t) = 1 + t + 5t^4$ is a polynomial of degree 4.

Transformations of functions

Vertical and Horizontal shifts

Suppose $c > 0$. To obtain the graph of

$y = f(x) + c$, shift the graph of $y = f(x)$ a distance c units upward

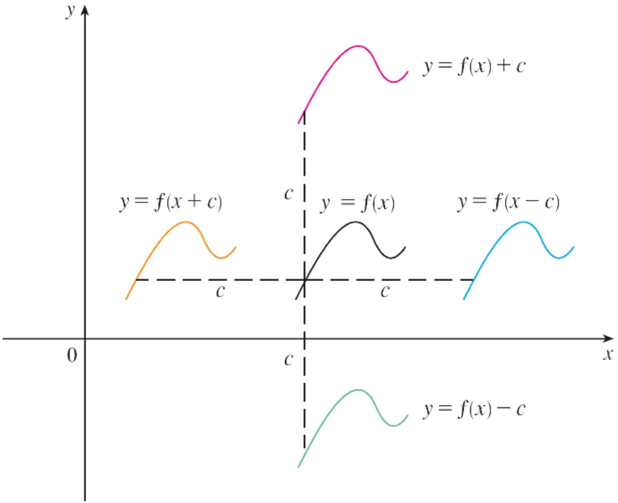
$y = f(x) - c$, shift the graph of $y = f(x)$ a distance c units downward

$y = f(x - c)$, shift the graph of $y = f(x)$ a distance c units to the right

$y = f(x + c)$, shift the graph of $y = f(x)$ a distance c units to the left

Transformations of functions

Vertical and Horizontal shifts



Transformations of functions

Vertical and Horizontal stretching and reflecting

Suppose $c > 1$. To obtain the graph of

$y = cf(x)$, stretch the graph of $y = f(x)$ vertically by a factor of c

$y = (1/c)f(x)$, shrink the graph of $y = f(x)$ vertically by a factor of c

$y = f(cx)$, shrink the graph of $y = f(x)$ horizontally by a factor of c

$y = f(x/c)$, stretch the graph of $y = f(x)$ horizontally by a factor of c

$y = -f(x)$, reflect the graph of $y = f(x)$ about the x-axis

$y = f(-x)$, reflect the graph of $y = f(x)$ about the y-axis

Vertical and Horizontal stretching and reflecting

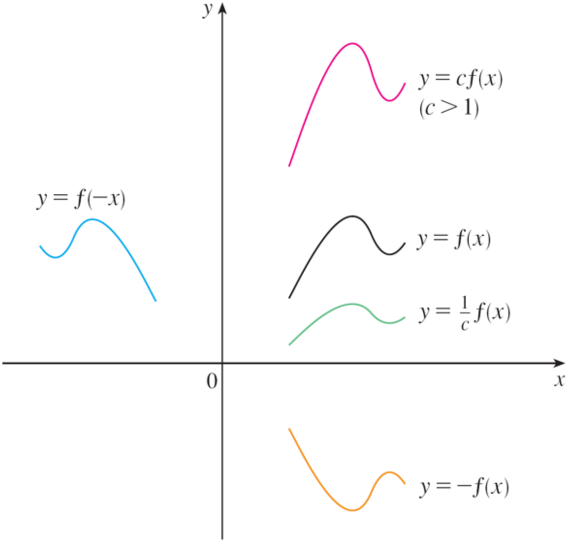
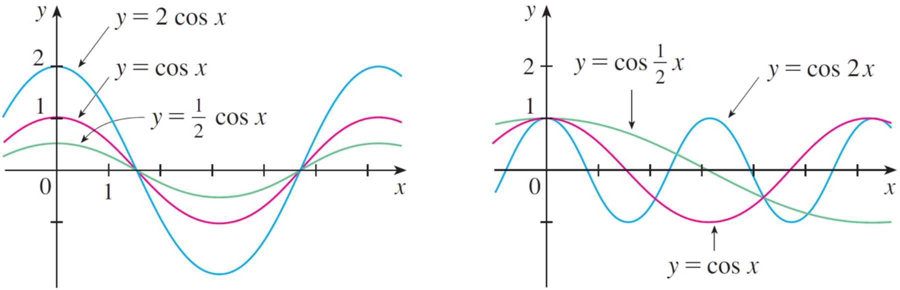


Figure below illustrates these stretching transformations when applied to the cosine function with $c = 2$.



Combinations of functions

Two functions f and g can be combined to form new functions $f+g$, $f-g$, fg, and f/g in a manner similar to the way we add, subtract, multiply, and divide real numbers.

The sum and difference functions are defined by

$$h(x) = \quad (f+g)(x) = f(x) + g(x) \qquad (f-g)(x) = f(x) - g(x)$$

If the domain of f is A and the domain of g is B, then the domain of $f \pm g$ is the intersection $A \cap B$ because both $f(x)$ and $g(x)$ have to be defined.

Similarly, the product and quotient functions are defined by

product function

$$(fg)(x) = f(x)g(x)$$

quotient function

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

The domain of fg is $A \cap B$. Because we cannot divide by 0, the domain of f/g is therefore $\{x \in A \cap B | g(x) \neq 0\}$.

Combinations of functions

Composite function

Given two functions f and g, the composite function $f \circ g$ (also called the composition of f and g) is defined by

$$(f \circ g)(x) = f(\overbrace{g(x)}^{\text{'}x\text{'}})$$ g(x) replaces the x in f(x)

The domain of $f \circ g$ is the set of all x in the domain of g such that $g(x)$ is in the domain of f. That is to say

for g(x) to be valid

$$\mathrm{Dom}(f \circ g) = \{\underline{x \in \mathrm{Dom}(g)} \,|\, \underbrace{g(x) \in \mathrm{Dom}(f)}_{\text{for } f(g(x)) \text{ is valid}}\}$$

It is possible to take the composition of three or more functions. For instance, the composite function $f \circ g \circ h$ is found by first applying h, then g, and then f as follows:

$$(f \circ g \circ h)(x) = f(g(h(x)))$$

Combinations of functions

Example C

Let $f(x) = \frac{x+1}{x+2}$ and $g(x) = x + \frac{1}{x}$. Find the function $f \circ g$ and its domain.

Solution We have

$\frac{g(x)+1}{g(x)+2}$

not the same function ∴ they have different domains

this excludes $x \neq 0$

$$(f \circ g)(x) = f(g(x)) = f(x + \frac{1}{x}) = \frac{x + \frac{1}{x} + 1}{x + \frac{1}{x} + 2}. = \frac{x^2+x+1}{x^2+2x+1} \Rightarrow \text{Domain: } \{x \in \mathbb{R} \mid x \neq -1\}$$

only when $x \neq 0$

Its domain is that $D = \{x \in \mathbb{R} | \underline{x \neq 0}, x + \frac{1}{x} + 2 \neq 0\} = \mathbb{R} - \{-1, 0\}$.

$x + \frac{1}{x} + 2 \neq 0$
$x + \frac{1}{x} \neq -2$
$x^2 + 1 \neq -2x$
$x^2 + 2x + 1 \neq 0$
$(x+1)^2 \neq 0$
$x + 1 \neq 0$
$x \neq -1$

Alternatively, to find the domain of $f \circ g$, note that Dom(f)=$\mathbb{R} - \{-2\}$ and Dom(g)=$\mathbb{R} - \{0\}$. The rule we will apply is that

$$\text{Dom}(f \circ g) = \{x \in \text{Dom}(g) | g(x) \in \text{Dom}(f)\}$$

Examining this rule we see that we just need to start with Dom(g) and remove from it those $x \in$ Dom(g) such that $g(x) \notin$ Dom(f), i.e. x such that $g(x) = -2$. The points we need to remove from Dom(g) are the x such that $\underline{g(x) = x + 1/x = -2}$. This equation has one solution: $\underline{x = -1}$. In conclusion: Dom$(f \circ g)$ = Dom$(g) - \{-1\} = \mathbb{R} - \{-1, 0\}$.

Exercise

Consider the function

$f(1)=1$
$f(0)=1$

$$f(x) = \begin{cases} 1, & \text{if } |x| \leq 1 \\ 0, & \text{if } |x| > 1 \end{cases}$$

Find $f\{f[f(x)]\}$.

$$f[f(x)] = \begin{cases} f(1), & \text{if } |x| \leq 1 \\ f(0), & \text{if } |x| > 1 \end{cases}$$
$$= 1$$

$$f\{f[f(x)]\} = f(1)$$
$$= 1$$

Combinations of functions

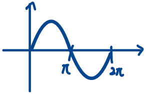
Exercise

The sign function of real number x is defined as follows:

$$\operatorname{sgn}(x) = \begin{cases} -1, & \text{if } x < 0 \\ 0, & \text{if } x = 0 \\ 1, & \text{if } x > 0 \end{cases}$$

Plot the graph of $\operatorname{sgn}(\sin x)$.

$$\sin x = \begin{cases} >0 , & 2k\pi < x < \pi + 2k\pi \\ <0 , & 2k\pi + \pi < x < 2k\pi + 2\pi \\ 0 , & x = k\pi \end{cases}$$



$$\operatorname{sgn}(\sin x) = \begin{cases} 1 , & 2k\pi < x < x + 2k\pi \\ -1 , & 2k\pi + \pi < x < 2k\pi + 2\pi \\ 0 , & x = k\pi \end{cases}$$

$k \in \mathbb{Z}$

Exercise

Given

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$

and

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta,$$

show that

$$\cos\alpha\sin\beta = \frac{1}{2}\left[\sin(\alpha+\beta) - \sin(\alpha-\beta)\right]$$

and

$$\sin^2\alpha = \frac{1}{2}\left[1 - \cos(2\alpha)\right].$$

$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

$\alpha = \beta$

$\Rightarrow \cos^2\alpha - \sin^2\alpha = \cos(2\alpha)$

$1-\sin^2\alpha - \sin^2\alpha = \cos(2\alpha)$

$2\sin^2\alpha = 1-\cos 2\alpha$

$\sin^2\alpha = \frac{1}{2}(1-\cos 2\alpha)$

$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

$\therefore \cos\alpha\sin\beta = \sin(\alpha+\beta) - \sin\alpha\cos\beta$

$\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$

$\therefore \cos\alpha\sin\beta = \sin\alpha\cos\beta - \sin(\alpha-\beta)$

$2\cos\alpha\sin\beta = \sin(\alpha+\beta) - \sin\alpha\cos\beta + \sin\alpha\cos\beta - \sin(\alpha-\beta)$

$\cos\alpha\sin\beta = \frac{1}{2}\left[\sin(\alpha+\beta) - \sin(\alpha-\beta)\right]$

Combinations of functions

Exercise

Given

$$\sinh(\alpha \pm \beta) = \sinh\alpha \cosh\beta \pm \cosh\alpha \sinh\beta$$

and

$$\cosh(\alpha \pm \beta) = \cosh\alpha \cosh\beta \pm \sinh\alpha \sinh\beta,$$

show that

$$\cosh\alpha \cosh\beta = \frac{1}{2}\left[\cosh(\alpha+\beta) + \cosh(\alpha-\beta)\right]$$

and

$$\sinh^2\alpha = \frac{1}{2}\left[\cosh(2\alpha) - 1\right].$$

$\cosh(\alpha+\beta) = \cosh\alpha\cosh\beta + \sinh\alpha\sinh\beta$
$\alpha = \beta$
$\cosh(2\alpha) = \cosh^2\alpha + \sinh^2\alpha$
$= 1 + \sinh^2\alpha + \sinh^2\alpha$
$2\sinh^2\alpha = \cosh(2\alpha) - 1$
$\sinh^2\alpha = \frac{1}{2}(\cosh 2\alpha - 1)$

$\cosh(\alpha+\beta) = \cosh\alpha\cosh\beta + \sinh\alpha\sinh\beta$
$\cosh\alpha\cosh\beta = \cosh(\alpha+\beta) - \sinh\alpha\sinh\beta$
$\cosh(\alpha-\beta) = \cosh\alpha\cosh\beta - \sinh\alpha\sinh\beta$
$\cosh\alpha\cosh\beta = \cosh(\alpha-\beta) + \sinh\alpha\sinh\beta$
$2\cosh\alpha\cosh\beta = \cosh(\alpha+\beta) + \cosh(\alpha-\beta)$
$\cosh\alpha\cosh\beta = \frac{1}{2}[\cosh(\alpha+\beta) + \cosh(\alpha-\beta)]$